Data Quality Audit Report

building_permits

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Data Health Score

Metric	Value
Total Rows	198,900
Total Columns	43
Health Score	73.6/100 (Good)
Critical Issues (High)	29
Warnings (Medium)	6
Info Items (Low)	17
Columns with Missing Data	31
Duplicate Rows	0

Executive Summary

This report presents a comprehensive data quality audit of **building_permits**, containing **198,900 rows** and **43 columns**. The overall data health score is **73.6/100** — rated **Good**.

A total of **29 high-severity**, **6 medium-severity**, and **17 low-severity** issues were identified across 5 quality-check categories.

Key findings: **TIDF Compliance** — 198898 missing values (100.0%); **Voluntary Soft-Story Retrofit** — 198865 missing values (100.0%); **Unit Suffix** — 196939 missing values (99.0%).

Immediate attention should be given to high-severity issues (missing values, duplicates, type inconsistencies) which have the greatest impact on analytical reliability. Detailed findings and prioritized recommendations follow.

Category	Weight	Score	Contribution
Completeness	35%	37.7	13.2
Uniqueness	15%	100.0	15.0
Type Consistency	20%	100.0	20.0
Outlier Reasonability	12%	91.8	11.0
Format Consistency	10%	75.0	7.5
Text Quality	8%	85.9	6.9
TOTAL	100%		73.6

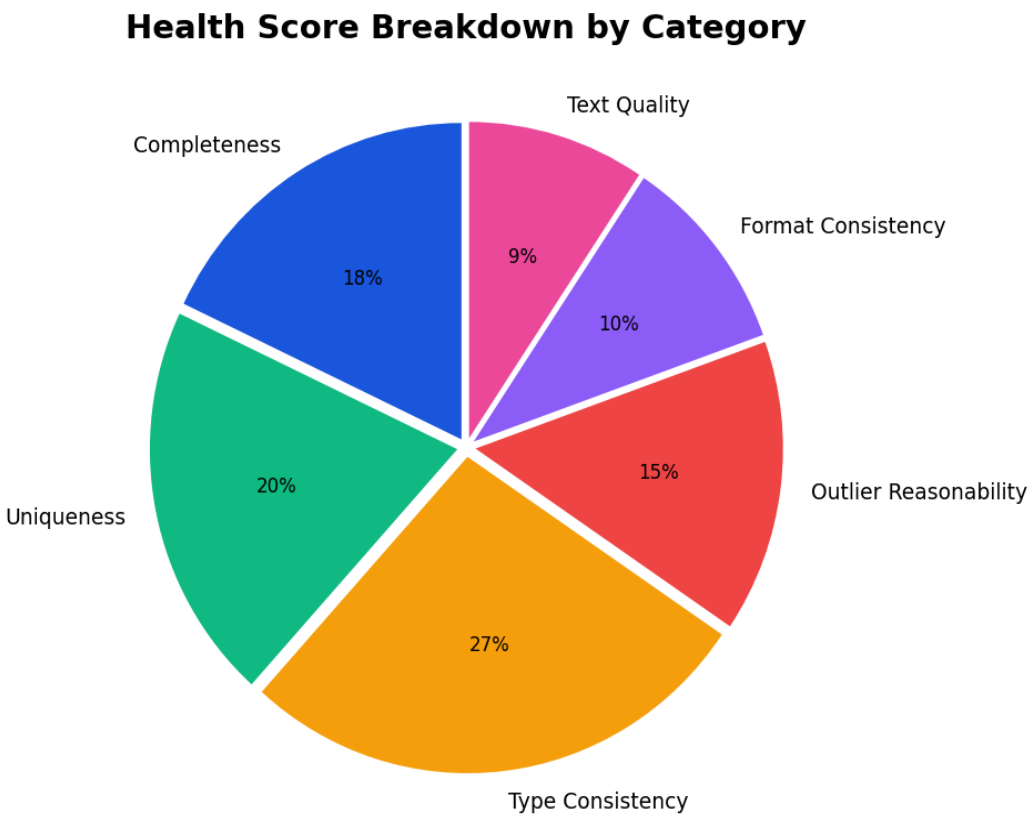

Figure 3: Health Score Breakdown by Quality Category

Findings Summary

The dataset **building_permits** contains **198,900 rows** and **43 columns**, with an overall data health score of **73.6/100**, rated **Good**.

The audit identified **52 issues** in total — **29 high-severity**, **6 medium-severity**, and **17 low-severity** — spanning 5 quality-check categories.

Completeness scored **37.7/100**: 31 of 43 columns contain missing values, with an average missing rate of **26.3%** across the dataset.

Columns **Street Number Suffix, Unit, Unit Suffix, Completed Date, Structural Notification, Number of Existing Stories, Number of Proposed Stories, Voluntary Soft-Story Retrofit, Fire Only Permit, Permit Expiration Date, Existing Use, Existing Units, Proposed Use, Proposed Units, TIDF Compliance, Existing Construction Type, Existing Construction Type Description, Proposed Construction Type, Proposed Construction Type Description, Site Permit** exceed the 20% missing threshold and require immediate imputation or removal.

Uniqueness scored **100.0/100**: no duplicate rows were detected.

Type Consistency scored a strong **100.0/100** with no type validation issues detected.

Format Consistency scored **75.0/100**: column **Permit Creation Date** contains 1 date format(s) detected: MM/DD/YYYY, which will cause parsing failures if not normalised to ISO 8601.

Text Quality scored **85.9/100**: inconsistent casing and/or leading/trailing whitespace were detected in **Description**.

1 column(s) have poor or auto-generated names (e.g. **Permit Number, Permit Type, Permit Type Definition, Permit Creation Date, Street Number, Street Number Suffix, Street Name, Street Suffix, Unit Suffix, Current Status, Current Status Date, Filed Date, Issued Date, Completed Date, First Construction Document Date, Structural Notification, Number of Existing Stories, Number of Proposed Stories, Voluntary Soft-Story Retrofit, Fire Only Permit, Permit Expiration Date, Estimated Cost, Revised Cost, Existing Use, Existing Units, Proposed Use, Proposed Units, TIDF Compliance, Existing Construction Type, Existing Construction Type Description, Proposed Construction Type, Proposed Construction Type Description, Site Permit, Supervisor District, Neighborhoods - Analysis Boundaries, Record ID**), which reduce maintainability and should be renamed or dropped.

Outlier Reasonability scored **91.8/100**: statistical outliers were found in 18 numeric column(s), with **Unit** being the most affected (21.1% of values outside IQR bounds).

The lowest-scoring quality dimension is **Completeness** at **37.7/100**, making it the primary focus area for remediation.

The top priority action is: **Impute (median/mode) or drop column if > 50% missing. 'TIDF Compliance' has 100.0% missing.**

Addressing all 29 high-severity issues is estimated to raise the health score significantly; detailed remediation steps are provided in the Recommendations section.

Missing Values Analysis

Of 43 columns, **31** contain missing values. Average missing rate: **26.3%**. Columns above 20% are flagged High and require imputation or removal.

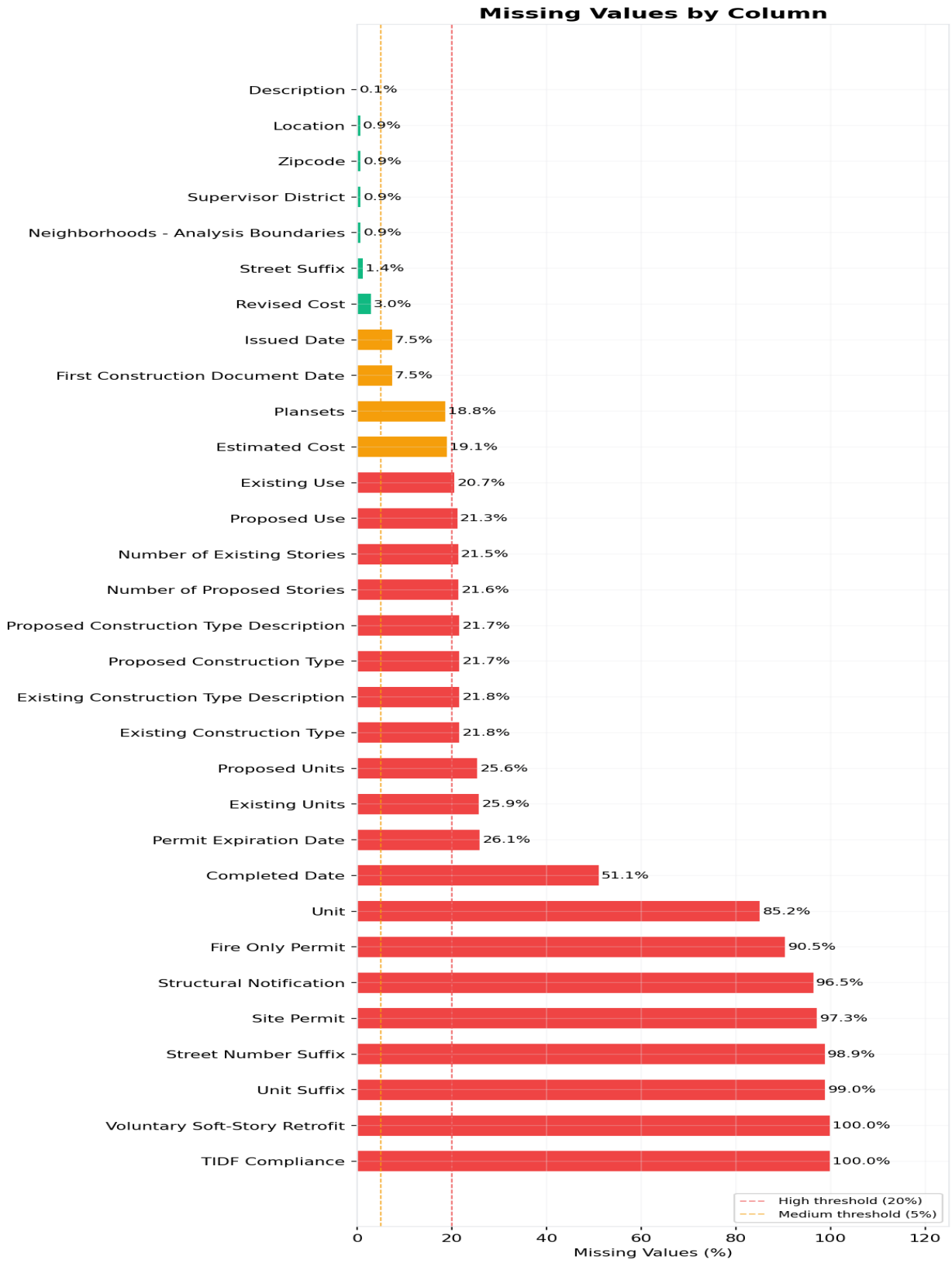

Figure 2: Missing values per column — red > 20%, orange 5–20%, green < 5%

Column	Missing Count	Missing %	Severity
TIDF Compliance	198898	100.0%	High
Voluntary Soft-Story Retrofit	198865	100.0%	High
Unit Suffix	196939	99.0%	High
Street Number Suffix	196684	98.9%	High
Site Permit	193541	97.3%	High
Structural Notification	191978	96.5%	High
Fire Only Permit	180073	90.5%	High
Unit	169421	85.2%	High
Completed Date	101709	51.1%	High
Permit Expiration Date	51880	26.1%	High
Existing Units	51538	25.9%	High
Proposed Units	50911	25.6%	High

Issue Severity Overview

The heatmap maps each quality check (rows) against each column (columns). **H** = High severity, **M** = Medium, **L** = Low, blank = no issue.

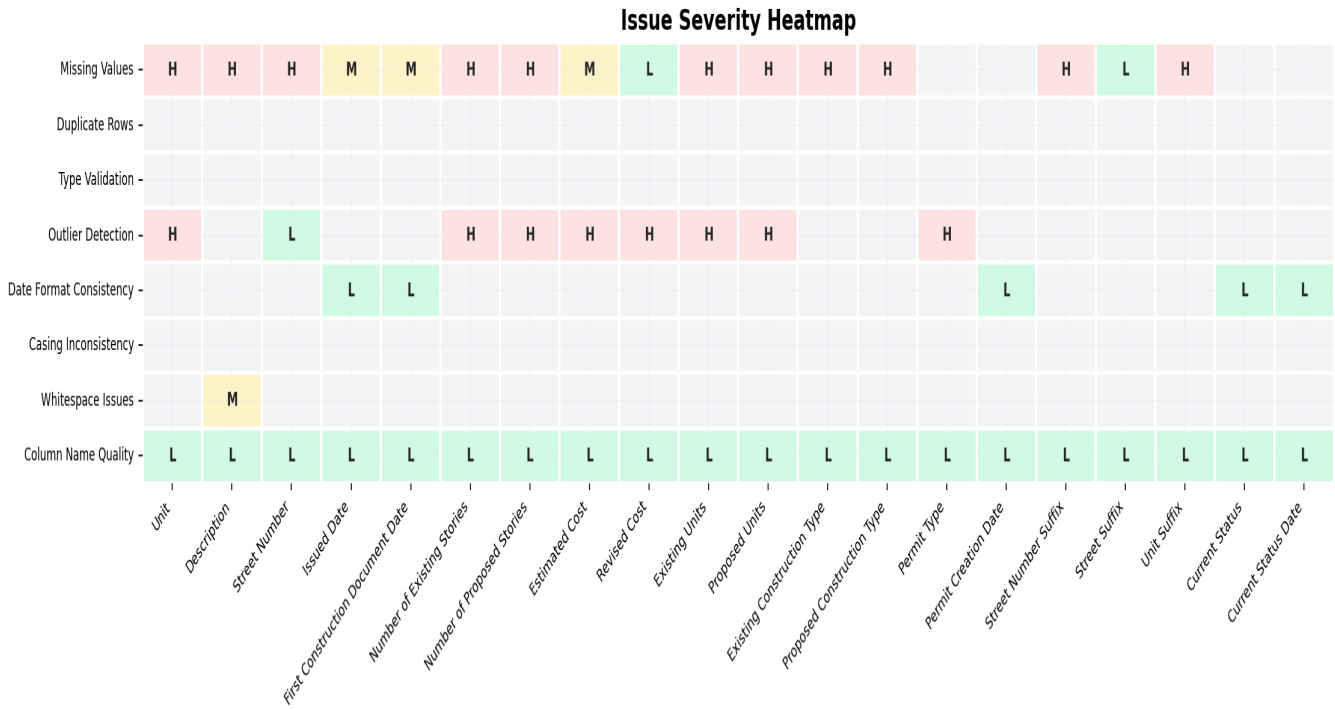

Figure 4: Issue Severity Heatmap

Top issue hotspots: **Unit** (score 22), **Description** (score 10), **Number of Existing Stories** (score 7), **Number of Proposed Stories** (score 7), **Existing Units** (score 7).

Distribution Analysis

Histograms for 6 numeric column(s). Red dashed = median; orange dotted = IQR outlier boundaries (Q1 – 1.5×IQR, Q3 + 1.5×IQR). Values beyond orange lines are statistical outliers.

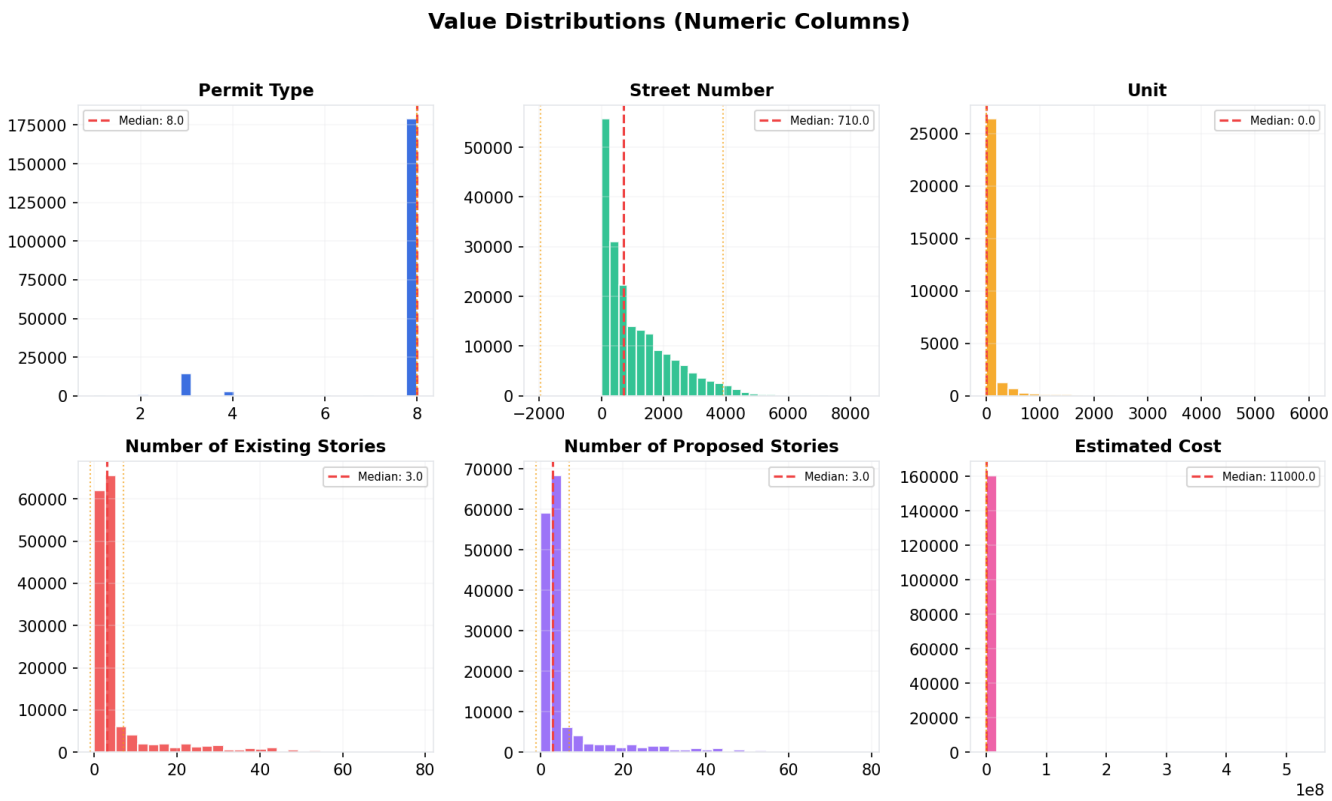

Figure 5: Value distributions for numeric columns

Permit Type: 20056 outlier(s) (10.1%), valid range [8.00, 8.00].

Street Number: 5853 outlier(s) (2.9%), valid range [-1962.50, 3897.50].

Unit: 6208 outlier(s) (21.1%), valid range [-1.50, 2.50].

Number of Existing Stories: 22495 outlier(s) (14.4%), valid range [-1.00, 7.00].

Number of Proposed Stories: 22423 outlier(s) (14.4%), valid range [-1.00, 7.00].

Estimated Cost: 20708 outlier(s) (12.9%), valid range [-44250.00, 82550.00].

Revised Cost: 25201 outlier(s) (13.1%), valid range [-43058.75, 71767.25].

Existing Units: 23784 outlier(s) (16.1%), valid range [-3.50, 8.50].

Proposed Units: 25042 outlier(s) (16.9%), valid range [-3.50, 8.50].

Plansets: 4 outlier(s) (0.0%), valid range [-3.00, 5.00].

Zipcode: 1058 outlier(s) (0.5%), valid range [94089.50, 94141.50].

Record ID: 36905 outlier(s) (18.6%), valid range [1118917412621.88, 1624648938606.88].

Block: 988 outlier(s) (0.5%), valid range [-3600.00, 8144.00].

Lot: 11507 outlier(s) (6.3%), valid range [-33.00, 79.00].

Trend Analysis

Numeric values over time using **Permit Creation Date**. Raw data shown at low opacity; solid line = rolling average. Note: dates were parsed from mixed formats (MM/DD/YYYY, YYYY-MM-DD, DD-MM-YYYY).

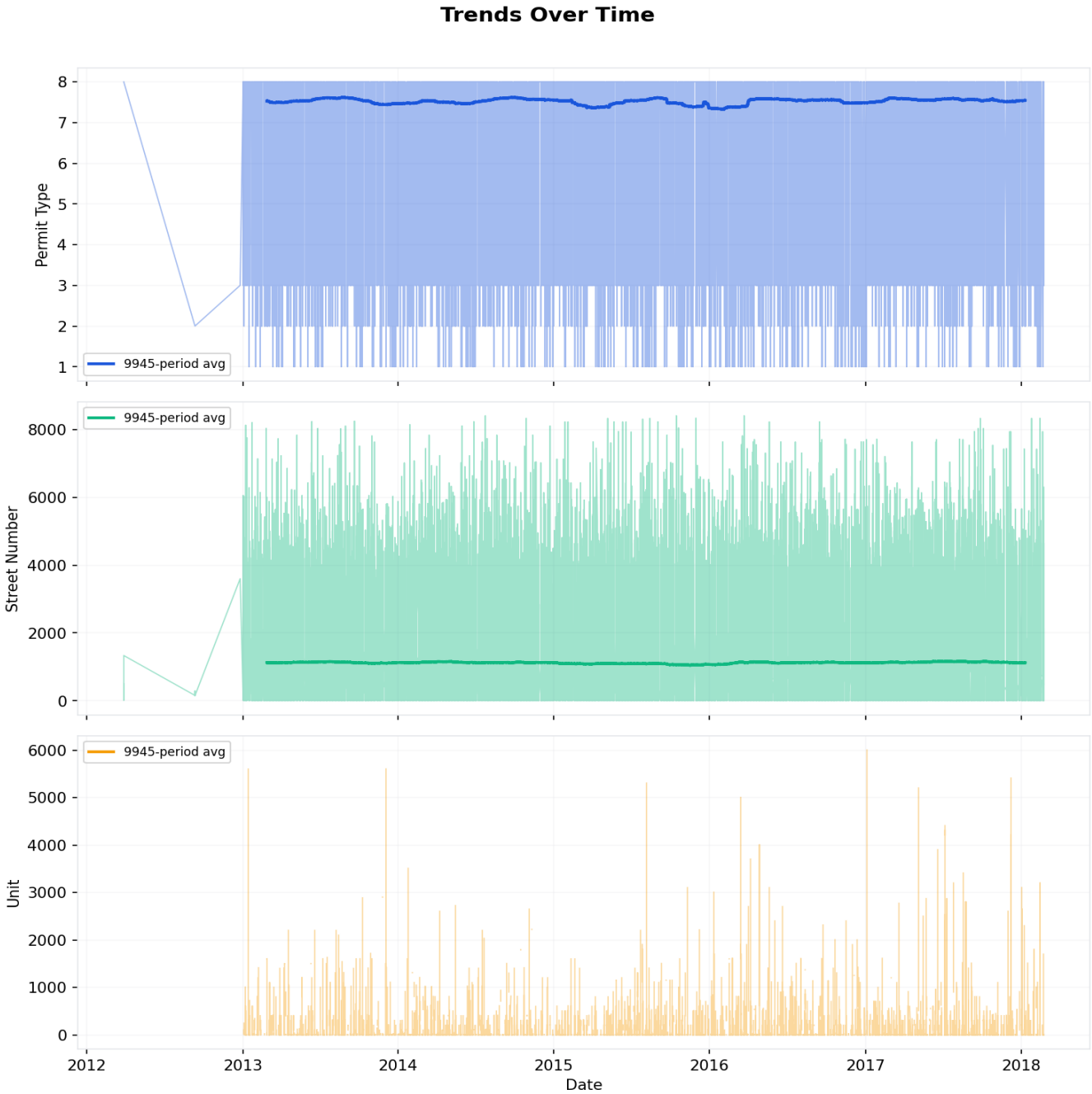

Figure 6: Trends Over Time with rolling averages

Detailed Findings

All 52 issues sorted by severity (High → Low), then affected rows.

Column	Check Type	Severity	Finding	Rows	%
TIDF Compliance	Missing Values	High	198898 missing values (100.0%)	198898	100.0%
Voluntary Soft-Story R	Missing Values	High	198865 missing values (100.0%)	198865	100.0%
Unit Suffix	Missing Values	High	196939 missing values (99.0%)	196939	99.0%
Street Number Suffix	Missing Values	High	196684 missing values (98.9%)	196684	98.9%
Site Permit	Missing Values	High	193541 missing values (97.3%)	193541	97.3%
Structural Notificatio	Missing Values	High	191978 missing values (96.5%)	191978	96.5%
Fire Only Permit	Missing Values	High	180073 missing values (90.5%)	180073	90.5%
Unit	Missing Values	High	169421 missing values (85.2%)	169421	85.2%
Completed Date	Missing Values	High	101709 missing values (51.1%)	101709	51.1%
Permit Expiration Date	Missing Values	High	51880 missing values (26.1%)	51880	26.1%
Existing Units	Missing Values	High	51538 missing values (25.9%)	51538	25.9%
Proposed Units	Missing Values	High	50911 missing values (25.6%)	50911	25.6%
Existing Construction	Missing Values	High	43366 missing values (21.8%)	43366	21.8%
Existing Construction	Missing Values	High	43366 missing values (21.8%)	43366	21.8%
Proposed Construction	Missing Values	High	43162 missing values (21.7%)	43162	21.7%
Proposed Construction	Missing Values	High	43162 missing values (21.7%)	43162	21.7%
Number of Proposed Sto	Missing Values	High	42868 missing values (21.6%)	42868	21.6%
Number of Existing Sto	Missing Values	High	42784 missing values (21.5%)	42784	21.5%
Proposed Use	Missing Values	High	42439 missing values (21.3%)	42439	21.3%
Existing Use	Missing Values	High	41114 missing values (20.7%)	41114	20.7%
Record ID	Outlier Detection	High	36905 outliers (18.6%), valid range [111891741262138,	36905	18.6%
Revised Cost	Outlier Detection	High	25201 outliers (13.1%), valid range [-43058.75, 717625.	25201	13.1%
Proposed Units	Outlier Detection	High	25042 outliers (16.9%), valid range [-3.50, 8.50]	25042	16.9%
Existing Units	Outlier Detection	High	23784 outliers (16.1%), valid range [-3.50, 8.50]	23784	16.1%
Number of Existing Sto	Outlier Detection	High	22495 outliers (14.4%), valid range [-1.00, 7.00]	22495	14.4%
Number of Proposed Sto	Outlier Detection	High	22423 outliers (14.4%), valid range [-1.00, 7.00]	22423	14.4%
Estimated Cost	Outlier Detection	High	20708 outliers (12.9%), valid range [-44250.00, 825500.	20708	12.9%
Permit Type	Outlier Detection	High	20056 outliers (10.1%), valid range [8.00, 8.00]	20056	10.1%
Unit	Outlier Detection	High	6208 outliers (21.1%), valid range [-1.50, 2.50]	6208	21.1%

Column	Check Type	Severity	Finding	Rows	%
Estimated Cost	Missing Values	Medium	38066 missing values (19.1%)	38066	19.1%
Plansets	Missing Values	Medium	37309 missing values (18.8%)	37309	18.8%
Description	Whitespace Issues	Medium	28124 values with leading/trailing or double spaces (28	28124	14.1%
First Construction Doc	Missing Values	Medium	14946 missing values (7.5%)	14946	7.5%
Issued Date	Missing Values	Medium	14940 missing values (7.5%)	14940	7.5%
Lot	Outlier Detection	Medium	11507 outliers (6.3%), valid range [-33.00, 79.00]	11507	6.3%
Permit Creation Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	198900	100.0%
Current Status Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	198900	100.0%
Filed Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	198900	100.0%
Issued Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	198900	100.0%
First Construction Doc	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	198900	100.0%
Permit Number, Permit	Column Name Quality	Low	36 column(s) with spaces/special chars in name	198900	100.0%
Revised Cost	Missing Values	Low	6066 missing values (3.0%)	6066	3.0%
Street Number	Outlier Detection	Low	5853 outliers (2.9%), valid range [-1962.50, 3897.50]	5853	2.9%
Street Suffix	Missing Values	Low	2768 missing values (1.4%)	2768	1.4%
Neighborhoods - Analys	Missing Values	Low	1725 missing values (0.9%)	1725	0.9%
Supervisor District	Missing Values	Low	1717 missing values (0.9%)	1717	0.9%
Zipcode	Missing Values	Low	1716 missing values (0.9%)	1716	0.9%
Location	Missing Values	Low	1700 missing values (0.9%)	1700	0.8%
Zipcode	Outlier Detection	Low	1058 outliers (0.5%), valid range [94089.50, 94141.50]	1058	0.5%
Block	Outlier Detection	Low	988 outliers (0.5%), valid range [-3600.00, 8144.00]	988	0.5%
Description	Missing Values	Low	290 missing values (0.1%)	290	0.1%
Plansets	Outlier Detection	Low	4 outliers (0.0%), valid range [-3.00, 5.00]	4	0.0%

Recommendations

Priority	Issue	Recommendation	Affected Columns	Est. Impact
P1 — Critical	Missing values in TIDF Compliance	Impute (median/mode) or drop column if > 50% missing. 'TIDF Compliance' has 100.0% missing.	TIDF Compliance	+30.0 pts on Completeness
P1 — Critical	Missing values in Voluntary Soft-Story Retrofit	Impute (median/mode) or drop column if > 50% missing. 'Voluntary Soft-Story Retrofit' has 100.0% missing.	Voluntary Soft-Story Retrofit	+30.0 pts on Completeness
P1 — Critical	Missing values in Unit Suffix	Impute (median/mode) or drop column if > 50% missing. 'Unit Suffix' has 99.0% missing.	Unit Suffix	+29.7 pts on Completeness
P1 — Critical	Missing values in Street Number Suffix	Impute (median/mode) or drop column if > 50% missing. 'Street Number Suffix' has 98.9% missing.	Street Number Suffix	+29.7 pts on Completeness
P1 — Critical	Missing values in Site Permit	Impute (median/mode) or drop column if > 50% missing. 'Site Permit' has 97.3% missing.	Site Permit	+29.2 pts on Completeness
P1 — Critical	Missing values in Structural Notification	Impute (median/mode) or drop column if > 50% missing. 'Structural Notification' has 96.5% missing.	Structural Notification	+29.0 pts on Completeness
P1 — Critical	Missing values in Fire Only Permit	Impute (median/mode) or drop column if > 50% missing. 'Fire Only Permit' has 90.5% missing.	Fire Only Permit	+27.2 pts on Completeness
P1 — Critical	Missing values in Unit	Impute (median/mode) or drop column if > 50% missing. 'Unit' has 85.2% missing.	Unit	+25.6 pts on Completeness
P1 — Critical	Missing values in Completed Date	Impute (median/mode) or drop column if > 50% missing. 'Completed Date' has 51.1% missing.	Completed Date	+15.3 pts on Completeness
P1 — Critical	Missing values in Permit Expiration Date	Impute (median/mode) or drop column if > 50% missing. 'Permit Expiration Date' has 26.1% missing.	Permit Expiration Date	+7.8 pts on Completeness
P1 — Critical	Missing values in Existing Units	Impute (median/mode) or drop column if > 50% missing. 'Existing Units' has 25.9% missing.	Existing Units	+7.8 pts on Completeness
P1 — Critical	Missing values in Proposed Units	Impute (median/mode) or drop column if > 50% missing. 'Proposed Units' has 25.6% missing.	Proposed Units	+7.7 pts on Completeness
P1 — Critical	Missing values in Existing Construction Type	Impute (median/mode) or drop column if > 50% missing. 'Existing Construction Type' has 21.8% missing.	Existing Construction Type	+6.5 pts on Completeness
P1 — Critical	Missing values in Existing Construction Type Description	Impute (median/mode) or drop column if > 50% missing. 'Existing Construction Type Description' has 21.8% missing.	Existing Construction Type Description	+6.5 pts on Completeness

Priority	Issue	Recommendation	Affected Columns	Est. Impact
P1 — Critical	Missing values in Proposed Construction Type	Impute (median/mode) or drop column if > 50% missing. 'Proposed Construction Type' has 21.7% missing.	Proposed Construction Type	+6.5 pts on Completeness
P1 — Critical	Missing values in Proposed Construction Type Description	Impute (median/mode) or drop column if > 50% missing. 'Proposed Construction Type Description' has 21.7% missing.	Proposed Construction Type Description	+6.5 pts on Completeness
P1 — Critical	Missing values in Number of Proposed Stories	Impute (median/mode) or drop column if > 50% missing. 'Number of Proposed Stories' has 21.6% missing.	Number of Proposed Stories	+6.5 pts on Completeness
P1 — Critical	Missing values in Number of Existing Stories	Impute (median/mode) or drop column if > 50% missing. 'Number of Existing Stories' has 21.5% missing.	Number of Existing Stories	+6.5 pts on Completeness
P1 — Critical	Missing values in Proposed Use	Impute (median/mode) or drop column if > 50% missing. 'Proposed Use' has 21.3% missing.	Proposed Use	+6.4 pts on Completeness
P1 — Critical	Missing values in Existing Use	Impute (median/mode) or drop column if > 50% missing. 'Existing Use' has 20.7% missing.	Existing Use	+6.2 pts on Completeness
P2 — High	Leading/trailing whitespace	Apply .str.strip() across all string columns. Use .str.replace(r'\s+', ' ', regex=True) for double spaces.	Description	+3 pts on Text Quality
P3 — Medium	Outliers in numeric columns (worst: Unit)	Investigate outlier rows — validate against source data. Use IQR capping (Winsorization) for extreme values. Do NOT blindly drop — confirm business context first.	Permit Type, Street Number, Unit	+1.2 pts on Outlier Reasonability